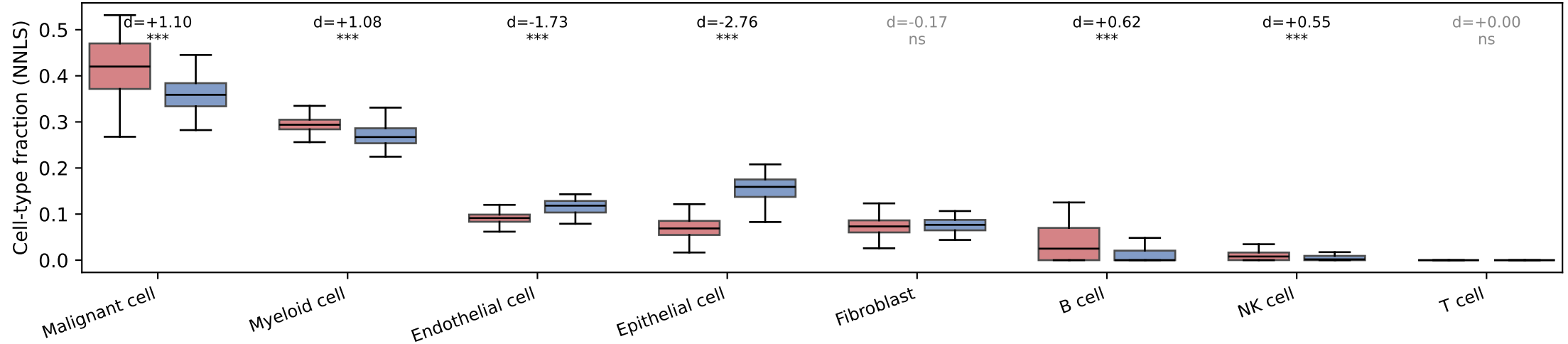
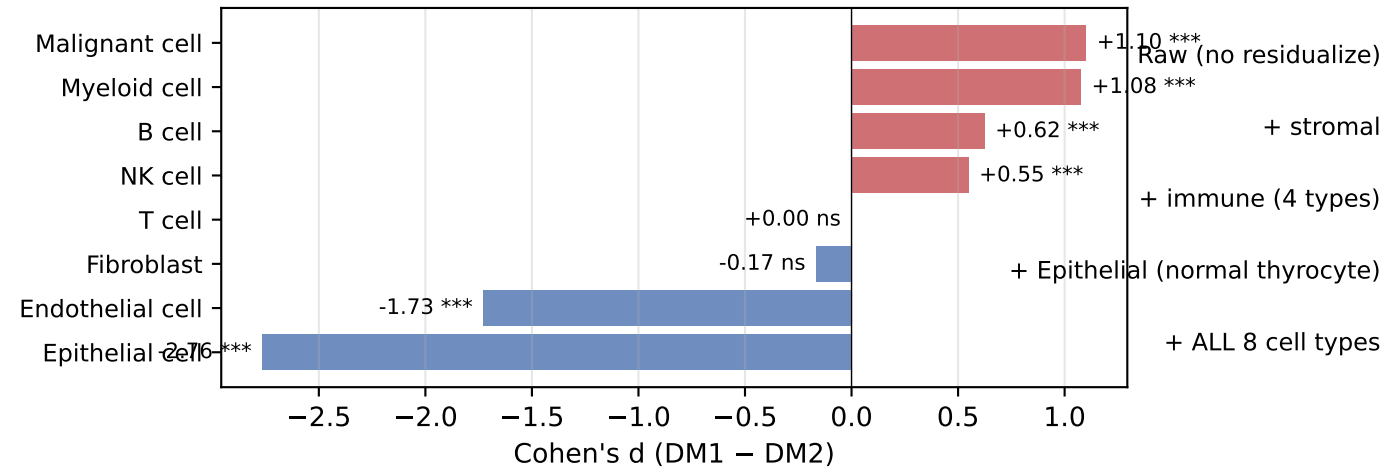


Supplementary Figure SX (deconvolution): bulk cell-type composition + residualization ladder Paper 1 manuscript v8 — Q10/Q11 reviewer-attack defense

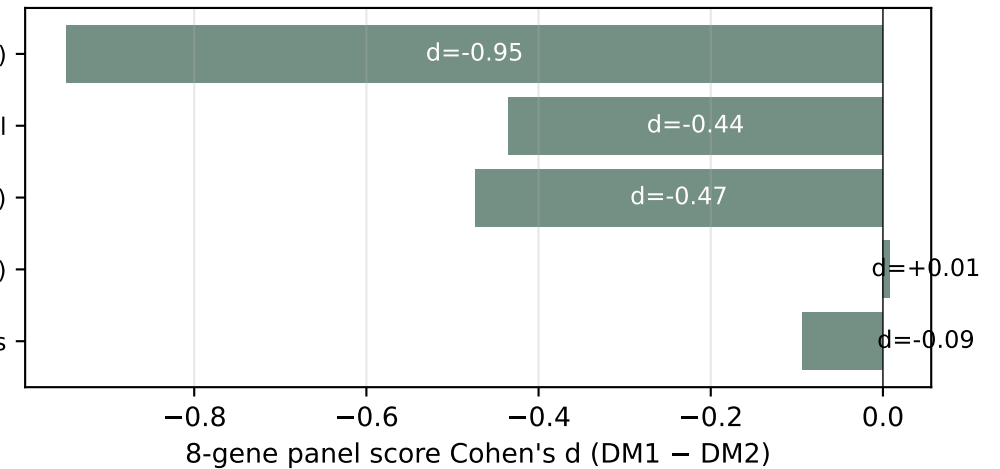
(A) Per-sample cell-type fractions: DM1 (red) vs DM2 (blue) — TCGA-THCA n=157



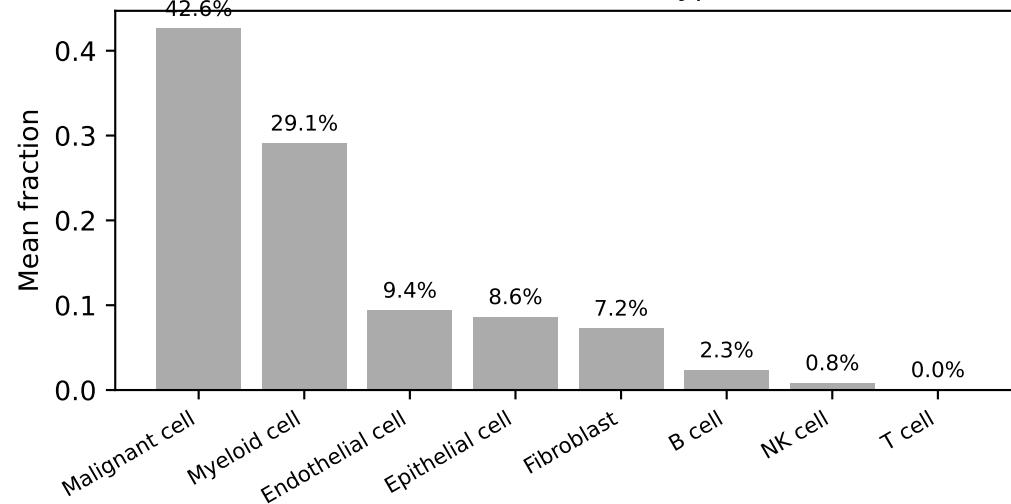
(B) DM1 vs DM2 cell-type fraction differences



(C) Residualization ladder — 8-gene panel signal



(D) TCGA-THCA n=572 mean cell-type fraction (NNLS)



INTERPRETATION & CAVEATS

Methodology: NNLS pseudobulk deconvolution
Reference: Lu 2023 (GSE193581) author_celltype, n=67,678 cells, 8 types
Bulk: TCGA-THCA n=572 (intersected to 1,898 HVG)
DM labels: dark_matter_phase2/p2d (DM1 n=96, DM2 n=61)

Key findings:

- DM1 has more Myeloid (d=+1.08), B cell (+0.62), NK (+0.55), Malignant (+1.10) vs DM2
- DM1 has less Epithelial cell (d=-2.76; normal thyrocyte admixture) and Endothelial (d=-1.73) vs DM2
- Raw 8-gene panel DM1 vs DM2 d=-0.95
- After residualization on Epithelial cell: d=+0.01 (signal collapses)
- After residualization on ALL 8 fractions: d=-0.09

Caveats:

- T cell fraction = 0 (NNLS collapse artifact); future work: ridge or DWLS
- Lu 2023 'Epithelial cell' (n=706) ≈ benign/normal thyroid epithelium; its DM1 vs DM2 difference reflects sample purity, NOT a cancer mechanism
- Per-patient sc analysis (Pu 2021, r=0.798-0.886 tumor-normal thyrocyte) addresses purity confounding by working WITHIN matched thyrocyte clusters